The conjectures of the revised proposal (chemistry prior, commitment, several dates): proofs, counterexamples and corrected statements

Second companion note

31 October 2026

Abstract

The revised proposal keeps the benchmark model (M2) and adds two relaxations: a two-class chemistry prior (M3) and commitment (M4). It states five conjectures. Conjectures 4 and 5, and the benchmark part of Conjecture 1, are unchanged; the first companion note settles them, and its results stand. This note settles the rest.

- *Conjecture 1 on the chemistry prior is false*, for two separate reasons. The after-date decision can make information pivotal, as in the benchmark. More strikingly, under any of the three stings, threshold rules fail once a relationship is valuable enough. Two mild conditions are needed: the low-chemistry chance of a yes clears the sting threshold, and the step is worth taking even with a low-chemistry match. A refusal is most informative at intermediate beliefs, so the agent asks when fairly sure the chemistry is low, and when fairly sure it is high, but not in between. Without a sting, and with an after-date plan that does not depend on beliefs, threshold rules hold at every step.
- *Conjecture 2 is true at certainty and false in general.* With a known high-chemistry match, a higher relationship value raises the gain from escalating. At a fixed belief short of certainty it can lower it, when commitment is weak relative to the fear of bad news; an exact asymptotic slope says where. The claim that the set of beliefs with negative commitment value shrinks with R holds wherever the after-date plan does not switch with the step reached. Where it does switch, the set can grow, in measure and in its upper end.
- *Conjecture 3 is false* in the several-dates model built on the chemistry prior, solved by value iteration: a better outside option can lower the threshold for continuing at beliefs where the agent would ask for another date. Its Phase C part is true.

Two further results bear on the proposal's open points. In the proposal's own regime, where refusals are bad news, information about chemistry has non-negative value at the last ask, under all three stings. And the proposal's Proposition 6 is true in M3. Every number is reproduced by `verify_v2.py` (model code in `chem.py`).

Summary

Claim	Verdict	What is proved
Conjecture 1, chemistry prior	False (Example 4, Proposition 5)	Thresholds in π at every step without a sting, when the after-date plan is the same at every belief (Theorem 3). With a sting, the last-step rule is "ask iff $A \geq \rho(\pi)$", with ρ zero at $\pi = 0, 1$ and positive in between. For large R this yields two asking bands, near 0 and near 1 (Proposition 5).
Conjecture 1, benchmark	Unchanged	First note: false in general, true under condition (M).
Conjecture 2, first part	True at $\pi = 1$; false at fixed $\pi < 1$ in general	At $\pi = 1$ the gain from a yes is nondecreasing in R, under any stings (Theorem 7). For large R, $\partial D/\partial R = \Phi(\pi)$ exactly, with $\Phi > 0$ near $\pi = 1$ and possibly $\Phi < 0$ elsewhere (Proposition 8).
Conjecture 2, second part	True without plan switches; false in general	The negative-commitment set shrinks with R at beliefs where the after-date plan does not switch, and always at $\pi = 0, 1$. With a switch it can grow, by inclusion, in measure and in its upper end, and it need not be an interval (Proposition 10).
Conjecture 3 (several dates, on the chemistry prior)	Phase C true; on the day false	Full-recursion counterexample by value iteration, with no sting, robust to the grid (Proposition 12).
Conjectures 4, 5	Unchanged	First note's results stand. The threshold part of Conjecture 4 extends to M3/M4 when refusals are bad news. The spend part of Conjecture 5 holds at the last ask under the chemistry prior in the proposal's regime (Section 5).

1 Setting

Notation follows the revised proposal. There are two classes, $z \in \{L, H\}$, with $(C, q, q_D) = (C_z, q_z, q_{D,z})$, $C_H > C_L$, $q_H > q_L$ and $q_{D,H} > q_{D,L}$ (A13). The belief is $\pi = P(z = H)$, and π_z denotes the weight on class z. At step j write

$$\bar{q} = \pi q_H + (1 - \pi) q_L, \qquad p = \varphi_j \bar{q}, \qquad \pi_{\text{yes}} = \frac{\pi q_H}{\bar{q}}, \qquad \pi_{\text{no}} = \frac{\pi(1 - \varphi_j q_H)}{1 - p}.$$

A short calculation gives

$$\pi_{\text{yes}} - \pi_{\text{no}} = \frac{\pi(1 - \pi)(q_H - q_L)}{\bar{q}\,(1 - \varphi_j \bar{q})}. \tag{1}$$

The effective weight is $\hat{p}(p) = p - \kappa_m(p)$, as in the first note.

Standing assumption S1. Ratings after step j are $\text{Bin}(m_j, C_z)$ given the class, with exact Bayes updating; this is MLRP in the class. The value after the date is truncated after one more date (21), except in Section 4. The relationship chance is $h_z(n)$ after a date that reached step n: $h_z(n) = h_z$ in M3, $\psi_n h_z$ (bonding) or $h_z + \theta_n$ (sliding) in M4, with $h_z = \bar{h}\, C_z q_{D,z}$. Another date has a fixed chance h_z^D, with $h_z^D = h_z$ in M3. The expected sting is $\kappa_m(p)\Delta^+$. This convention departs from the proposal's maintained $\Delta > 0$ only when $\Delta \leq 0$. It affects no result below; Section 5 notes where it would matter. The Phase-C sting is written κ_C when it is distinguished from the on-day sting.

Standing assumption S2. Where stated, the ask for another date carries no sting. This is the condition of the proposal's Propositions 4–8.

For a Phase-C plan $d \in \{\text{cut}, \text{ask}\}$, let X_z^d be its value against class z:

$$X_z^{\text{cut}} = U, \qquad X_z^{\text{ask}} = q_{D,z} V_{D,z} + (1 - q_{D,z})U, \qquad V_{D,z} = \delta_D\big(h_z^D C_z R + (1 - h_z^D)U\big).$$

Lemma 1 (After-date value)**.** *Under S1 and S2,*

$$V_{\text{end}}(\pi; n) = \max_d \sum_z \pi_z \Big[h_z(n) C_z R + \big(1 - h_z(n)\big) X_z^d \Big].$$

This is convex and piecewise linear in π. It is affine wherever the optimal plan is the same at every belief.

Proof. Bayes' rule after "no relationship" gives $(1 - \mathbb{E}h)\, f(\pi_{\text{not}}) = \sum_z \pi_z (1 - h_z(n)) f(\delta_z)$ for every f that is linear in the belief. The value of each plan is linear in the belief once the sting is absent, so $(1 - \mathbb{E}h) V_C(\pi_{\text{not}}) = \max_d \sum_z \pi_z (1 - h_z(n)) X_z^d$. □

Lemma 2 (Anatomy of an ask)**.** *At step j, with n steps completed, let X be the value after a yes (step, rating, continuation), and let*

$$G = X - V_{\text{end}}(\pi; n), \qquad L = V_{\text{end}}(\pi; n) - V_{\text{end}}(\pi_{\text{no}}; n), \qquad \Delta = G + L.$$

The value of asking minus ending is $D = pG - (1 - p)L - \kappa(p)\Delta^+$. Suppose $V_{\text{end}}(\cdot; n)$ is affine with slope a_1, and write

$$A = X - V_{\text{end}}(\pi_{\text{yes}}; n), \qquad \Gamma = a_1(\pi_{\text{yes}} - \pi_{\text{no}}).$$

Then

$$D = p\, A - \kappa(p)\, (A + \Gamma)^+. \tag{2}$$

Proof. Bayes plausibility gives $p\pi_{\text{yes}} + (1 - p)\pi_{\text{no}} = \pi$, so an affine V_{end} satisfies $pV_{\text{end}}(\pi_{\text{yes}}) + (1 - p)V_{\text{end}}(\pi_{\text{no}}) = V_{\text{end}}(\pi)$. Hence $pG - (1 - p)L = pA$. Also $\Delta = X - V_{\text{end}}(\pi_{\text{no}}) = A + \Gamma$. □

Here A is the value of the step net of the news a yes carries, and Γ is the gap the news opens between a yes and a no. By (1), Γ is zero at $\pi = 0$ and $\pi = 1$. When $a_1 > 0$, so that refusals are bad news, it is positive in between.

2 Conjecture 1 on the chemistry prior

Theorem 3 (Thresholds without a sting)**.** *Under S1 and S2 in M3, suppose there is no sting on the day ($\eta = 0$) and the optimal Phase-C plan is the same at every π. Then for every step, $D_j = p_j A_j$ with A_j nondecreasing in π. So the agent asks exactly on $\{\pi \geq \pi_j^*\}$.*

Proof. By Lemma 1, V_{end} is affine, and Lemma 2 with $\kappa = 0$ gives $D_j = p_j A_j$. Since the rating is a martingale on the affine V_{end},

$$A_j(\pi) = W_j^*(\pi_{\text{yes}}) - c_j + \mathbb{E}\big[O_{j+1}(\pi') \mid \pi_{\text{yes}}\big], \qquad O_{j+1} = D_{j+1}^+.$$

By induction, $O_{j+1} = (p_{j+1} A_{j+1})^+$ is nondecreasing: p is increasing and A_{j+1} is nondecreasing. The posterior after a rating is increasing in the prior and in the rating, and the rating's predictive distribution rises with the belief, so the expectation is nondecreasing in π_{yes}. W_j^* is increasing in $\mathbb{E}[C]$. Finally, π_{yes} is increasing in π. Since $p_j > 0$, D_j has the sign of A_j. □

Example 4 (Counterexample A: the after-date decision is pivotal)**.** Take M3 with $C = (0.2, 0.8)$, $q = (0.895, 0.9)$, $q_D = (0.795, 0.8)$, $\bar{h} = 0.5$, $R = 6000$, $U = 1000$ and $\delta_D = 0.9$. The last step has $v = 50$ and $k = 30$ with one Bernoulli rating, and there is no sting anywhere. The agent asks on $[0.0681, 0.2340] \cup [0.6654, 1]$ (computed). The mechanism is that of the first note's counterexample to Conjecture 1, carried into M3.

Proposition 5 (Counterexample B: fear of finding out)**.** *Under S1 and S2 in M3, take the last step ($O \equiv 0$), and suppose the Phase-C plan is constant, so V_{end} is affine with slope $a_1 > 0$. Then $A(\pi) = W^*(\pi_{\text{yes}}) - c$ is nondecreasing, and the agent asks if and only if*

$$\hat{p}(p) > 0 \quad \textit{and} \quad A(\pi) \;\geq\; \rho(\pi) := \frac{\kappa(p)}{\hat{p}(p)}\, a_1\, \frac{\pi(1-\pi)(q_H - q_L)}{\bar{q}(1-\varphi\bar{q})}. \tag{3}$$

Suppose further:

- *$\kappa > 0$ on $(0,1)$: D1 or D2 with $\eta > 0$, or KR with $\eta(\Lambda - 1) > 0$;*
- *$\hat{p}(\varphi q_L) > 0$;*
- *the step is worth taking even with a low-chemistry match: $A(0) = W^*(C_L) - c > 0$.*

Then there is $\bar{a}$ such that, whenever $a_1 > \bar{a}$, the agent asks for all π near 0 and all π near 1, but not at $\pi = 1/2$. In particular, if $\delta_D = 0$, or if $\delta_D h_L^D C_L > 0$, then $a_1 \to \infty$ as $R \to \infty$, so the asking set fails to be a threshold for every R large enough.

Proof. The rule (3). In (2), if $A + \Gamma > 0$ then $D = \hat{p}A - \kappa\Gamma$, which is non-negative iff $A \geq \kappa\Gamma/\hat{p}$ when $\hat{p} > 0$. If $A + \Gamma \leq 0$ then $D = pA \leq 0$. If $\hat{p} \leq 0$ and $A + \Gamma > 0$ then $D < 0$, because $\Gamma > 0$. Substitute (1).

Near 0 *and* 1. As $\pi \to 0$, we have $\rho \to 0$, $A \to A(0) > 0$ and $\hat{p} \to \hat{p}(\varphi q_L) > 0$, so $D > 0$ near 0. On $\{\hat{p} > 0\}$, $\hat{p}$ is increasing for all three models. Under D2 and KR write $\hat{p} = p(1 - \eta' + \eta' p)$, with $\eta' = \eta$ (D2) or $\eta(\Lambda - 1)$ (KR); its derivative $1 - \eta' + 2\eta' p$ is positive when $\eta' \leq 1$, and when $p > 1 - 1/\eta'$ otherwise. So $\hat{p}(\varphi q_H) > 0$. Also $A(1) \geq A(0) > 0$, and $\rho \to 0$ as $\pi \to 1$, so $D > 0$ near 1.

At 1/2. $A(1/2)$ does not depend on a_1, while $\rho(1/2) = a_1 \cdot c_0$ with $c_0 > 0$. So $D(1/2) < 0$ once $a_1 > \bar{a} := \max\{0, A(1/2)/c_0\}$.

Large R. If $V_{D,L} > U$, which holds for large R when $\delta_D h_L^D C_L > 0$, both classes favour asking, so the plan is "ask" at every belief. If $\delta_D = 0$ and $U \geq 0$ it is "cut" everywhere. Then

$$a_1 = R\,(\bar{c}_H - \bar{c}_L) + \text{const},$$

where, under "ask", $\bar{c}_z = h_z C_z + (1 - h_z) b_z$ with $b_z = q_{D,z}\delta_D h_z^D C_z$; under "cut", $b_z = 0$. Writing $\bar{c}_z = b_z + h_z(C_z - b_z)$, with $b_z \leq C_z$, shows that $\bar{c}_z$ is increasing in h_z, b_z and C_z, all of which are larger for $z = H$. So $\bar{c}_H > \bar{c}_L$. □

Example 6. Take $C = (0.5, 0.6)$, $q = (0.5, 0.95)$, $q_D = (0.5, 0.9)$, $\bar{h} = 1$, $U = 100$ and $\delta_D = 0.9$. The last step has $v = 40$, $k = 10$ and one rating.

- D1 with $\eta = 0.5$, $R = 1000$: the agent asks on $[0, 0.0404] \cup [0.6904, 1]$.
- Same, $R = 3000$: the agent asks on $[0, 0.0103] \cup [0.9003, 1]$.
- D2 with $\eta = 0.8$, or KR with $\eta(\Lambda - 1) = 0.8$, $R = 3000$: the agent asks on $[0, 0.0154] \cup [0.9328, 1]$.

The Phase-C plan is "ask" at every belief. Without a sting the agent asks everywhere.

The intuition is that a refusal tells the agent most when they are most unsure. Near $\pi = 0$ a no confirms what they expected; near $\pi = 1$ it is unlikely and barely moves them. In between, a no is likely enough and informative enough that, with a large stake, its expected sting outweighs a step that is worth having for its own sake. This is a new mechanism; it does not arise in the benchmark, where refusals carry no news.

For the guide, a single threshold score per step (Table 5 of the proposal) is not enough. Under any sting the card may need a low asking band and a high one. Both shrink towards the ends as R grows, consistent with Proposition 7 of the proposal. M4 nests M3 ($\psi_n \equiv 1$, $\theta_n \equiv 0$), so both counterexamples apply there too.

3 Conjecture 2: relationship value with commitment

Theorem 7 (First part, at certainty)**.** *Take M4 under S1, with $h_H(n)$ nondecreasing in n. Stings are arbitrary, on the day and in Phase C, and the budget is arbitrary. At $\pi = 1$, the gain from a yes, G_j, is nondecreasing in R at every step and every budget. If $\hat{p}_j > 0$, the value of escalating D_j is also nondecreasing, and the set of R at which a known high-chemistry match is asked to escalate is an upper set. If $\hat{p}_j \le 0$, the agent never asks at step j.*

Proof. At $\pi = 1$ nothing is learned, $p = \varphi_j q_H$, and a no ends the date at the step already reached, so $L = 0$. In Phase C the plan maximises the value against class H,

$$X_H(R) = U + \big(q_{D,H} - \kappa_C(q_{D,H})\big)^+ \big(V_{D,H}(R) - U\big)^+,$$

which does not depend on n. Hence

$$V_{\text{end}}(1; m; R) = X_H(R) + h_H(m)\big(C_H R - X_H(R)\big).$$

X_H is convex and piecewise linear with slope at most $\delta_D h_H^D C_H \le C_H$, so $C_H R - X_H(R)$ is nondecreasing. So $V_{\text{end}}(1; m; \cdot)$ has increasing differences in (m, R).

As in the first note's lemma that refusals carry no news, $V_j(1; n; B) = V_{\text{end}}(1; n) + \hat{p}_j^+ G_j(B)^+$, with

$$G_j(B) = \sup_{s \in S_j(B)} \Big\{ W_j(1; s) - c_j + \big[V_{\text{end}}(1; n+1) - V_{\text{end}}(1; n)\big] + \hat{p}_{j+1}^+ \, G_{j+1}\big(B - \sigma(k_j + s)\big)^+ \Big\}.$$

For each s the bracket is nondecreasing in R, and so is $G_{j+1}(B')$, for every B', by induction over j; the flows do not depend on R. A supremum of nondecreasing functions is nondecreasing. The base case is the last step, where the last term is absent. Finally, $D = pG$ for $G < 0$ and $D = \hat{p}G$ for $G \ge 0$, which is nondecreasing in G when $\hat{p} \ge 0$. □

Proposition 8 (First part, short of certainty)**.** *Under S1 and S2, take the last step after n steps, a belief $\pi \in (0, 1]$, and suppose $\delta_D h_L^D C_L > 0$ (or $\delta_D = 0$). Let*

$$\bar{c}_z(m) = h_z(m) C_z + (1 - h_z(m)) \, q_{D,z} \, \delta_D \, h_z^D C_z, \qquad \bar{K}(\pi) = \sum_z (\pi_{\text{yes}})_z \big[\bar{c}_z(n+1) - \bar{c}_z(n)\big].$$

($\bar{c}_H(n) > \bar{c}_L(n)$ holds automatically, as in the proof of Proposition 5.) Then there is $\bar{R}(\pi)$ such that, for $R \ge \bar{R}$, $D(\pi; R)$ is affine in R with slope

$$\Phi(\pi) = \hat{p}(p) \, \bar{K}(\pi) \;-\; \kappa(p) \, (\pi_{\text{yes}} - \pi_{\text{no}}) \big[\bar{c}_H(n) - \bar{c}_L(n)\big]. \tag{4}$$

(i) $\Phi(1) = \hat{p}(\varphi q_H)\big[\bar{c}_H(n+1) - \bar{c}_H(n)\big]$, which is non-negative when $\hat{p}(\varphi q_H) \ge 0$. If it is positive, $\Phi > 0$ on a neighbourhood of $\pi = 1$, and Conjecture 2(a) holds there for all large R.

(ii) Where $\Phi(\pi) < 0$, a higher R lowers the value of escalating, and the agent stops asking for large R. Proposition 7 of the proposal then survives commitment.

Proof. For large R the plan is "ask" at every belief, as in Proposition 5. Then

$$V_{\text{end}}(\pi; m) = R \sum_z \pi_z \bar{c}_z(m) + \sum_z \pi_z e_z(m),$$

with e_z free of R. The rating is a martingale on this affine function, so

$$G = W^* - c + V_{\text{end}}(\pi_{\text{yes}}; n+1) - V_{\text{end}}(\pi; n), \qquad L = V_{\text{end}}(\pi; n) - V_{\text{end}}(\pi_{\text{no}}; n).$$

$\Delta = G + L$ has R-coefficient $\sum \pi_{\text{yes}} \bar{c}(n+1) - \sum \pi_{\text{no}} \bar{c}(n)$, which is positive, so $\Delta > 0$ for large R. $D = pG - (1-p)L - \kappa\Delta$ is then affine in R. Using $p\pi_{\text{yes}} + (1-p)\pi_{\text{no}} = \pi$, the slope of $pG - (1-p)L$ is $p\bar{K}$, and the slope of $\kappa\Delta$ is

$$\kappa\big[\bar{K} + (\pi_{\text{yes}} - \pi_{\text{no}})(\bar{c}_H(n) - \bar{c}_L(n))\big].$$

Subtracting gives (4). Continuity of Φ gives (i). □

Example 9. Take $C = (0.3, 0.8)$, $q = q_D = (0.5, 0.9)$, $\bar{h} = 0.5$, $\delta_D = 0.9$, D1 with $\eta = 0.5$, and a last step with $v = 40$ and $k = 15$.

- Weak bonding, $\psi = 1.05$: at $\pi = 0.9$, $D = 16.81,\ 13.43,\ 0.74,\ -16.17$ for $R = 10^3,\ 5 \cdot 10^3,\ 2 \cdot 10^4,\ 4 \cdot 10^4$, and $\Phi(0.9) = -0.0008$. At $\pi = 1$, D rises: 28.3, 63.0, 193.1, 366.5.
- Strong bonding, $\psi = 1.5$: $\Phi(0.9) = 0.068 > 0$.

With $\psi = 1.05$ the region where $\Phi < 0$ ends near $\pi \approx 0.91$, so $\pi = 0.9$ lies just inside it. The numerical slopes equal (4) to six decimals.

So "likely high-chemistry" must be quantified. A higher R helps escalation exactly where $\Phi > 0$, that is, where commitment outweighs the sting of the news a refusal carries. That region always contains a neighbourhood of certainty, and can be small.

Proposition 10 (Second part)**.** *Take sliding, S1 and S2, let $K(\pi; R) = V_{\text{end}}(\pi; n+1) - V_{\text{end}}(\pi; n)$, and let $\Delta\theta = \theta_{n+1} - \theta_n > 0$.*

(i) Suppose at π the optimal plan is the same plan d at n and at $n+1$, for every R in $[R_1, R_2]$. Then

$$K(\pi; R) = \Delta\theta \sum_z \pi_z \big(C_z R - X_z^d(R)\big)$$

is nondecreasing on $[R_1, R_2]$. So if π has non-negative commitment value at R_1, it keeps it.

(ii) At $\pi = 0$ and $\pi = 1$ this holds for all R. If there is no second-date option (plan "cut"), the negative set is

$$N(R) = \Big[0,\ \frac{U - C_L R}{(C_H - C_L)R}\Big),$$

which shrinks as R rises.

(iii) In general the conjecture is false.

- *Take $C = (0.28, 0.93)$, $q_D = (0.71, 0.99)$, $\bar{h} = 0.85$, $\delta_D = 0.95$, $U = 100$ and $\theta_{n+1} - \theta_n = 0.06$. At $\pi = 0.27$, $K = +0.271$ at $R = 230$ but $K = -0.362$ at $R = 250$ (exact in rational arithmetic). Here $N(R)$ is not an interval. Its measure and upper end still fall between these two values of R, so this example breaks only inclusion.*
- *Take $C = (0.1397, 0.8664)$, $q_D = (0.9160, 0.9619)$, $\bar{h} = 0.9541$, $\delta_D = 0.9617$, $U = 100$ and $\theta_{n+1} - \theta_n = 0.1159$. From $R = 147.44$ to $R = 169.39$, the measure of $N(R)$ rises from 0.741 to 0.798, and its upper end from 0.741 to 0.801. At $\pi = 0.77$, K goes from $+0.154$ to -0.290. This breaks the conjecture under every reading of "the range shrinks".*

Proof. (i) By Lemma 1, with the plan fixed, K is as displayed. Moreover $\partial X_z^{\text{ask}} / \partial R = q_{D,z} \delta_D h_z^D C_z \le C_z$, and $X_z^{\text{cut}} = U$.

(ii) At a degenerate belief, π_{not} is the same degenerate belief, so the plan maximises X_z^d and does not depend on n. Then

$$C_z R - \max_d X_z^d = \min_d \big(C_z R - X_z^d\big)$$

is nondecreasing.

(iii) Computed in `verify_v2.py`. In the first example, at $R = 230$ the agent would ask for another date after a date that reached n steps, but cut after one that reached $n + 1$: sliding makes "no relationship" worse news about chemistry. At $R = 250$ the agent asks after both. Commitment then locks in a relationship that, at this belief, is worth less than the second-date plan ($\sum \pi_z (C_z R - X_z^{\text{ask}}) < 0$). □

Remark 11. In money, the values of K involved are small: below 0.3 on a scale where $U = 100$. In beliefs, though, the effect is not small. The upper end of the negative set can move by about 0.06 (and by 0.1 at intermediate R), and the violations scale with $\Delta\theta$. Conjecture 2(b) fails exactly where the after-date plan switches between n and $n + 1$.

4 Conjecture 3 in the several-dates model

Conjecture 3 is stated for M2+, the full date-by-date recursion. Here the recursion is built on the chemistry prior, which keeps the belief one-dimensional and the after-date update closed form; call it M3+. U is exogenous, as in the conjecture's comparative static. Talking between dates is switched off; take c_{talk} large. A date starts with its opening step, then asks; after it comes a relationship, or Phase C, where another date is worth $\delta_D V_{\text{date}}(\pi_{\text{yes}})$. The fixed point V_{date} is computed by value iteration on a grid of π; the contraction factor is at most δ_D.

Proposition 12. *(a)* Phase C (true). *With no sting anywhere, the set of beliefs at which the agent asks for another date shrinks as U rises.*

(b) On the day (false). *Take $C = (0.24, 0.63)$, $q = (0.74, 0.76)$, $q_D = (0.42, 0.59)$, $\bar{h} = 0.36$, $R = 1470$ and $\delta_D = 0.88$. The opening step is coffee ($v = 10$, $k = 8$, one rating); the second step has $v = 96$, $k = 50$ and two ratings. There is no sting.*

- *The on-day asking set is a threshold at both $U = 313$ and $U = 322$.*
- *Raising U* lowers *the threshold, from* 0.6272 *to* 0.6119. *At both thresholds the agent would ask for another date.*
- *The Phase-C asking region moves from $\pi \geq 0.385$ to $\pi \geq 0.406$, as part (a) says.*

The numbers are identical on grids of 401 *and* 1601 *points.*

Proof of (a). With no sting, every policy's value has the form $\alpha + \beta U$. Here $\beta \in [0, 1]$ is the expected discount factor at the time the agent returns to search, and zero if they never do. So $V_{\text{date}}(\pi; U)$ is a supremum of affine functions with slopes in $[0, 1]$. Hence $\delta_D V_{\text{date}}(\pi'; U) - U$ is strictly decreasing in U. In Phase C a no leads to U, so the agent asks iff $\delta_D V_{\text{date}}(\pi_{\text{yes}}) \geq U$. □

The mechanism is the pivotality of the first note, proved there for the truncated model and illustrated here in the full recursion. The value of learning on the day comes from the after-date decision, and a better outside option moves that decision towards indifference, which makes learning more valuable. The full recursion does not remove this. The version of M2+ with independent Beta priors is not covered, because "no relationship" breaks conjugacy there; the mechanism is the same.

5 Conjectures 4 and 5, and the value of information

The statements of Conjectures 4 and 5, and of Prediction 7 of Appendix A.9, are unchanged, and the first note's results apply. Two points are specific to the chemistry prior.

Conjecture 4. The first note shows that asking sets shrink as the budget falls in the core model. That carries over to M3 and M4 whenever refusals are weakly bad news ($L \geq 0$) at every reachable belief. The argument is by induction over the remaining steps. Suppose the value after a yes, X, is nondecreasing in the budget. If $\hat{p} \geq 0$, the value of asking, $pX + (1-p)V_{\text{end}}(\pi_{\text{no}}) - \kappa(X - V_{\text{end}}(\pi_{\text{no}}))^+$, is nondecreasing in X. If $\hat{p} < 0$, then $D = \hat{p}\Delta - L < 0$ when $\Delta > 0$, and $D = p\Delta - L \leq 0$ when $\Delta \leq 0$, so the agent never asks at any budget. Either way $V_j = \max\{V_{\text{end}}, \text{ask}\}$ is nondecreasing in the budget, which passes the hypothesis to the previous step.

Conjecture 5 and the value of information. The first note's guarantee that spend is never below Proposition 1's level rests on information about compatibility having non-negative value. Under the chemistry prior every signal about the class also moves the chance of a yes, so that argument has to be redone. At the last ask, it goes through in the proposal's regime.

Proposition 13 (Information has non-negative value at the last ask)**.** *Under S1 and S2 in M3, take the last ask, with a constant Phase-C plan and $a_1 \geq 0$, so that refusals are bad news. Assume $\hat{p}(\varphi q_L) \geq 0$. Assume also $\Delta \geq 0$ at every π (D1), or $A \geq 0$ at every π (D2, KR). Then V is convex in π, so every signal about the class has non-negative EVSI. Spend that sharpens the step's signal is therefore never below Proposition 1's level, by the argument of part (i) of the first note's spend-to-learn proposition.*

Proof. Write $t = \bar{q}$, which is affine in π, and $y = \pi_{\text{yes}}$, so $y' > 0$ and $y'' = -2t'y'/t$. With $\Delta \geq 0$, (2) gives $D = \hat{p}A - \kappa\Gamma$, with $A = W^*(y) - c$; W^* is C^1, convex and increasing.

- Under D1, $\hat{p} = (1+\eta)\varphi t - \eta$ and $\kappa\Gamma = \eta(1-p)\Gamma = \eta a_1(y - \pi)$, using $(1-p)(\pi_{\text{yes}} - \pi_{\text{no}}) = \pi_{\text{yes}} - \pi$. Then, wherever W^* is twice differentiable,

$$(\hat{p}A)'' = \hat{p}\, W^{*\prime\prime} y'^2 + 2\eta\, W^{*\prime}\, t'y'/t \;\geq\; 0.$$

  The term $-\eta a_1(y - \pi)$ is convex, because y is concave and $a_1 \geq 0$.

- Under D2 and KR, $\hat{p} = \varphi t(1 - \eta' + \eta'\varphi t)$, and $\kappa\Gamma = \eta'\varphi a_1 \pi(1-\pi)(q_H - q_L)$, which is concave. Then

$$(\hat{p}A)'' = 2\eta'\varphi^2 t'^2 A + \hat{p}\, W^{*\prime\prime} y'^2 + 2\eta'\varphi^2\, t\, W^{*\prime} t'y' \;\geq\; 0$$

  when $A \geq 0$.

So D is convex on $[0,1]$; at kinks of $W^{*\prime}$ it remains convex because $W^{*\prime}$ is continuous and nondecreasing. Hence $V = V_{\text{end}} + D^+$ is convex. Convexity of the value in beliefs is equivalent to non-negative EVSI for every signal. □

A random search of 582 last-step models satisfying the conditions found no failure of convexity. The reviewer's broader search, about 2.2 million M3 models with two or three steps and the after-date plan free, kept the cases with V_{end} increasing and $\Delta > 0$, $L \geq 0$ at the prior and both posteriors. Its minimum EVSI was zero up to round-off. Beyond the last ask the claim is therefore supported numerically but not proved.

Remark 14 (Where the Δ^+ convention matters). Negative EVSI can be produced when refusals are *good* news ($a_1 < 0$, which needs $C_H R < U$) and the stake changes sign between posteriors.

- Take $C = (0.11, 0.6)$, $q = (0.81, 0.88)$, $q_D = (0.5, 0.9)$, $\bar{h} = 0.5$, $R = 129$, $U = 100$, $\delta_D = 0$, a last step with $v = 31$ and $k = 10$, and D1 with $\eta = 1.5$.
- Here $a_1 = -3.74$. A cue with $P(\text{enc} \mid H) = 0.52$ and $P(\text{enc} \mid L) = 0.48$, observed at $\pi = 0.44$, has stakes -0.37, -0.07 and $+0.24$ at the low posterior, the prior and the high posterior.
- With $\kappa\Delta^+$ its EVSI is -0.0249. With the proposal's literal $\kappa\Delta$ it is $+0.0032$.

This lies outside the proposal's maintained $\Delta > 0$, and is an artefact of the convention rather than a finding about the model.

6 Other points on the revised proposal

1. *Proposition 6 holds in M3.* Here $h^D = h$ and $q_{D,H} > q_{D,L}$, and the proposal's hypothesis is that each plan pays at least as much against H as against L. Proof sketch (due to the reviewer): write $V_{D,z} - U = \delta_D h_z(C_z R - U) - (1 - \delta_D)U$.
   - If $V_{D,H} \leq U$, then $q_{D,H}(V_{D,H} - U) \geq q_{D,L}(V_{D,L} - U)$ with $q_{D,H} > q_{D,L}$ forces $V_{D,H} \geq V_{D,L}$.
   - If $V_{D,H} > U$, then $C_H R > U$.

Either way $h_H(C_H R - U) \geq h_L(C_L R - U)$, which is the slope of V_{end} at $\pi = 0$ under "cut". Under "ask" optimal at $\pi = 0$, $X_z^{\text{ask}} \leq C_z R$, and $X_H^{\text{ask}} \geq X_L^{\text{ask}}$ gives a non-negative slope. Convexity (Lemma 1) then makes V_{end} nondecreasing, so $L \geq 0$. A random check over about 6,000 models found no exception. The condition can fail if h_z^D is chosen unrelated to h_z.

2. *Table 3 and Prediction 6.* The benchmark's investment motive is not strongest "where beliefs are least settled". The first note shows it peaks near the after-date indifference point and can be zero at maximal uncertainty, which is also what the proposal's own Proposition 4 implies.

3. *Table 5.* "One threshold score per step" should become "up to two asking bands per step" wherever a sting is used (Proposition 5). When the no-sting policy asks everywhere, as in Example 6, the sting check of Section 6.1 flags exactly the middle band.

4. *Section 4.4.* It still lists the sign of the value of information as open, while Section 10.1 says the claim is withdrawn. For the benchmark, the first note's value-of-information theorem settles it. For the chemistry prior, Proposition 13 settles it at the last ask in the proposal's regime.

5. *Conjecture 2.* Restate "for likely high-chemistry matches" as "where $\Phi > 0$" (Proposition 8). Restate the second part as holding where the after-date plan does not switch with the step reached.

6. *Conjecture 3.* Keep the Phase C part. Replace the on-day part by the pivotality pattern: thresholds rise with U where the agent would cut contact, and can fall where they would ask for another date. This is proved in the truncated model (first note) and illustrated in the full recursion (Proposition 12).

Numerical verification

`python3 verify_v2.py` (about one minute) reproduces every number above. It also runs randomised checks of:

- Theorem 3, as single crossing at every step in 150 models;
- Theorem 7, in 150 M4 models with random stings, bonding or sliding;
- the rule (3) and formula (1);
- formula (4);
- Proposition 10(i)–(ii), over 115,000 parameter pairs;
- Proposition 12(a), in random models with the full recursion;
- Proposition 13, as convexity in 582 models;
- Proposition 6 of the proposal, in about 6,000 M3 models.

The output is in `verify_v2_output.txt`.